Regulatory Reform and the Composition of FDA Drug Approvals

Evidence on Controlled Substances, Regulatory Incentives, and Pharmaceutical Market Composition

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Did User Fees Change What the FDA Approves?

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- **Question:** Did PDUFA or GDUFA shift FDA approvals toward controlled substances?
- **Answer: No.** The hypothesis is not supported by the data.

What This Paper Adds

- First systematic test of whether PDUFA or GDUFA changed the *therapeutic composition* of FDA approvals.
- Two literatures—FDA user-fee effects (*Berndt et al., 2005; Olson, 2009; Philipson et al., 2008*) and the economics of opioid supply (*Alpert, Powell, Pacula, 2018*)—have not been joined.
- Paper provides the upstream, approval-side counterpart to downstream studies of opioid market dynamics.

FDA Drugs@FDA × DEA Scheduling, 1939–2025

- **25,908** original drug approvals (Drugs@FDA, March 2026)
- **2,067** confirmed controlled substances (8.0%), linked by ingredient to DEA schedules
- **86.6%** of CS approvals are generics (ANDAs)—NDA-only subsample is the policy-relevant unit for PDUFA

Era	Total	CS%
Pre-PDUFA ($\leq$1992)	7,301	9.6
PDUFA I–II (1993–02)	3,187	9.2
PDUFA III–IV (2003–12)	5,042	7.5
PDUFA V–VI (2013–22)	7,844	7.4
PDUFA VII (2023+)	2,534	4.5
Total	25,908	8.0

CS = confident DEA scheduled match (conservative)

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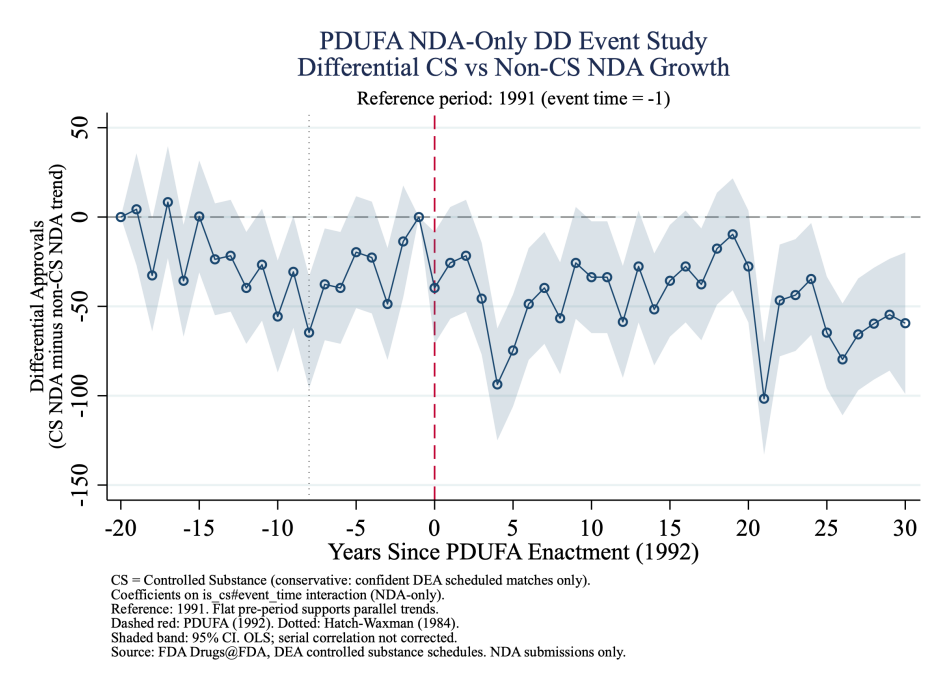
$$\log \mathbb{E}[n_{gt}] = \alpha_g + \lambda_t + \underbrace{\delta}_{\text{coeff. of interest}} \cdot (\mathrm{CS}_g \times \mathrm{Post}_t) + \log(\mathrm{Total}_t)$$

NDA-only sample for PDUFA; ANDA-only for GDUFA. Transition years 2013–2014 excluded.

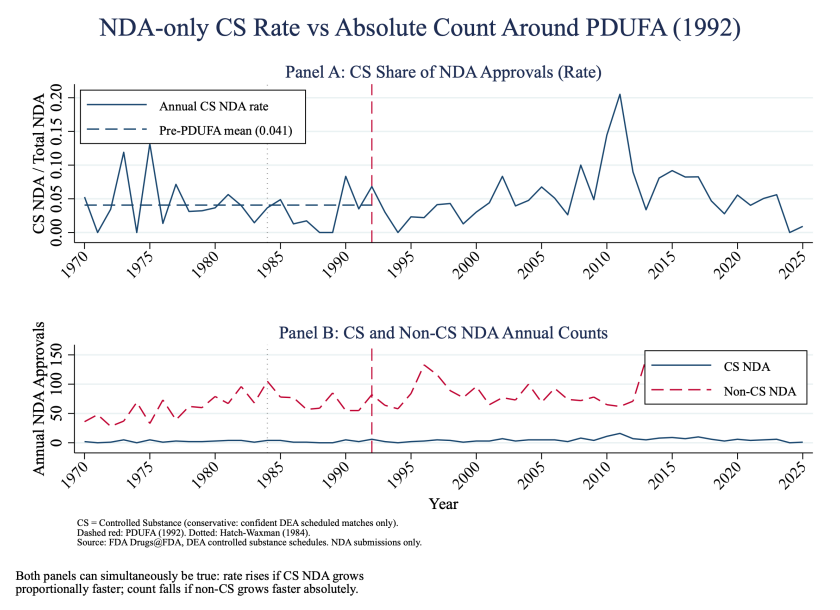
PDUFA: Marginally Significant, But Not Robust



- Pre-period broadly stable; some downward drift
- Post period: IRR $= 1.46$, $p = 0.06$
- IRR collapses to 1.19 ($p = 0.62$) with group-specific trends

Specification	IRR	p
Year FE	1.46	0.06
Linear trend	1.46	0.07
Group trends	**1.19**	**0.62**

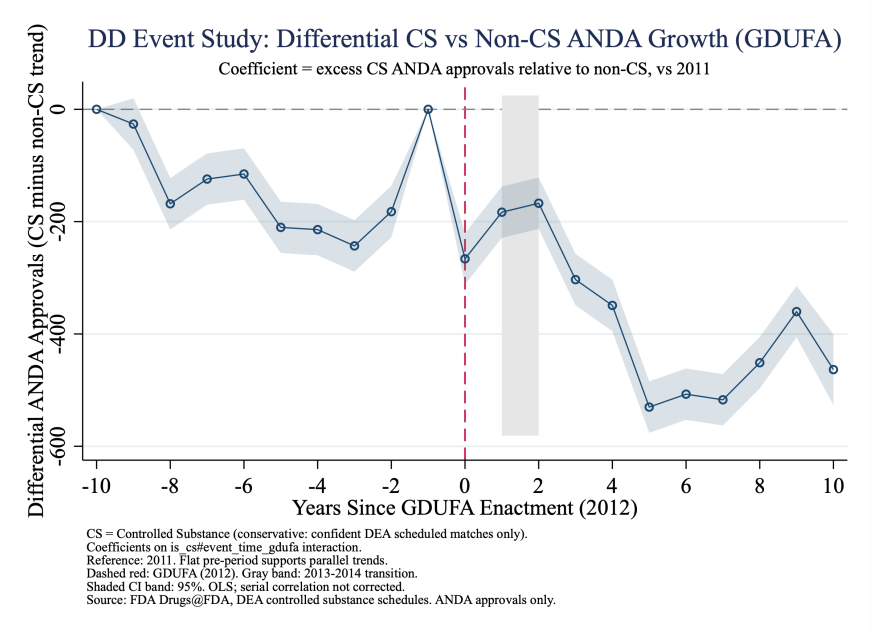
The "Effect" Is a 2009–2013 Opioid Product Cycle



Year	Drug	Sched.
2009	EMBEDA	II
2010	EXALGO	II
2010	OXYCONTIN	II
2010	SUBOXONE	III
2010	BUTRANS	III
2011	INTERMEZZO	IV
2012	QSYMIA	IV
2013	ZOHYDRO ER	II
43 CS NDAs; 58% Schedule II		

Timing is 17–19 years after PDUFA. Not a regulatory effect—an industry product cycle.

GDUFA: CS ANDA Share Declines, Not for User-Fee Reasons



- Year FE: IRR $= 0.668$, $p < 0.001$ —CS share falls 33%
- Group-specific trends: IRR $= 1.38$, $p = 0.18$ —same sensitivity as PDUFA
- Candidate drivers: pre-GDUFA backlog clearance, post-2012 opioid market contraction, patent-cliff ANDA surge

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- When apparent effects appear, they trace to the industry product cycle, not to regulatory incentives.
- **Policy channel matters:** the pathway from regulatory acceleration to CS supply runs through **firm innovation decisions**, not review-time margins.

Upstream approval-side evidence complements Alpert, Powell & Pacula (2018) and related downstream work.

What the Design Can and Cannot Show—And What's Next

Limitations

- Single national time series; no cross-sectional control
- DEA scheduling is current, not historical-at-approval
- Approval data only—failed applications unobserved

Future work

- Link CS approval counts to ARCOS/IQVIA prescribing data
- 2014 hydrocodone reschedule as a cleaner natural experiment
- Firm/therapeutic-class variation to identify mechanism
- User-fee dose-response analysis

Regulatory acceleration, on the margins of the approval pipeline, did not tilt the FDA toward controlled substances.